Mobile Visualisation Apps for In-Situ Museum Explorations



CS4099: Major Software Project

Christopher G Dove
110005609
4th April 2016

Supervisor: Dr Uta Hinrichs

Abstract

A museum is a rich, complex, information space. Every artefact is the centrepoint to a fascinating context that spans many dimensions, which could be historical, geographical or social, amongst others. In exploring a museum, a visitor may want to look deeper into one of these dimensions of an artifact's context or they may want to draw connections between different artifacts in order to find related artefacts and gain broader knowledge. Some museum visitors find that technology can assist or enhance these processes of deepening and broadening knowledge. This study aims to design, produce and evaluate a mobile visualisation app that will assist a museum visitor in in-situ exploration.

I declare that the material submitted for assessment is my own work except where credit is explicitly given to others by citation or acknowledgment. This work was performed during the current academic year except where otherwise stated. The main text of this project report is 7078 words long, including project specification and plan.

In submitting this project report to the University of St Andrews, I give permission for it to be made available for use in accordance with the regulations of the University Library. I also give permission for the title and abstract to be published and for copies of the report to be made and supplied at cost to any bona fide library or research worker, and to be made available on the World Wide Web. I retain the copyright in this work.

If there is a strong case for the protection of confidential data, the parts of the declaration giving permission for its use and publication may be omitted by prior permission of the Honours Coordinator.

Contents page

1. Introduction

2. Objectives

3. Context survey

3.1 Spatial layout of Musa

3.2 Technical provisions at Musa

4. Software engineering process

4.1 Functional requirements

4.2 Non-functional requirements

5. Ethics

6. Design

6.1 Sketch-based Ideation Process

6.2 Graph-based Interface

7. Implementation

7.1 Limitations of the implementation

8. Evaluation/Critical Assessment

8.1 Walkthrough at Musa

8.2 Adding Perspective

8.3 Force Directed Graph Prototype

8.4 Conclusions from the Evaluation

9. Conclusion

10. References

1. Introduction

Information Visualisation is a branch of computer graphics. In 2008 a study called EMDialog aimed to bring information visualisation to the museum space (Hinrichs, Schmidt & Carpendale, 2008). In 2014 a study called It Doesn't Compare to Being There (Rogers, Hinrichs & Quigley, 2016) involved the production of a visualisation for Musa, the Museum of the University of St Andrews. It was found that an in-situ visualisation is preferable to a remote one.

An app for in-situ museum exploration should take an item-centric approach to artefact selection. Often provided on a mobile device, it should provide the user with additional artefact information and assist the user in finding other artefacts that may be of interest to them. This functionality should be provided without distracting the user's attention from the physical space. Accessible to a wide demographic of museum visitors, the visualisation should be intuitive to use and should be visually attractive in order to appeal to that wider demographic.

The aim of this project is to produce an in-situ information visualisation for Musa that solves these problems. It also aims to evaluate the visualisation in order to see that it fulfills the requirements of Musa.

2. Objectives

Primary objectives

Design and implement a prototype for an information visualisation that enables museum visitors to explore individual museum artefacts and promote exploration of related ones. This will include:

- Ideation and the development of at least 3 low fidelity prototypes (sketches and paper)
- Research of related projects in this area
- Design and implementation of one idea into a stable high fidelity prototype based on a sample collection, which
    - Enables the selection of artefacts of interest (e.g. through QR codes or other methods).
    - Provides additional information about the selected artefact in visual and textual form.
    - Uses visual and interactive methods to suggest related artefacts and/or information in an intuitive, engaging and playful way.

Secondary objectives

- Evaluate the prototypes with curators and potential visitors, focusing on usability and user experience.

Tertiary objectives

- The implementation produced under the primary objectives would only incorporate a subset of the MUSA data. Expand the prototype to work with a larger subset or the entire collection.
- Expand the prototype to include participatory features such as letting the user submit comments on the artefacts.

3. Context survey

Prior to starting the project it was necessary to develop an awareness of the relevant technologies and research that may influence the direction of the work. In particular, it was necessary to gain awareness of the available visualisation tools and the devices that may support the visualisation. It was also necessary to survey recent research papers in the area of the work in order to develop well-informed functional requirements for the prototypes.

3.1 Spatial layout of Musa

The Musa museum space comprises 4 galleries.

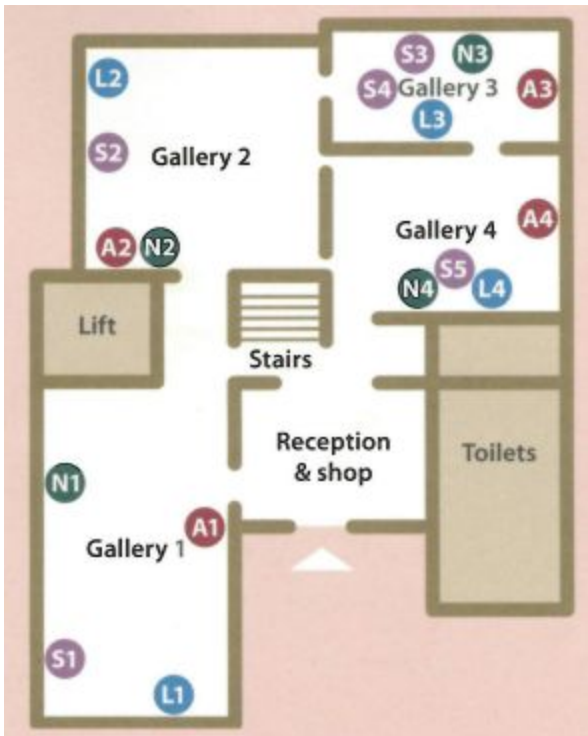


Figure 1. Map of the Musa Galleries (*Musa Visitor Guide*)

The Musa museum space comprises 4 galleries. The galleries are numbered Gallery 1, Gallery 2, Gallery 3 and Gallery 4, respectively, as labelled in figure 1. Each gallery has a corresponding name. The gallery called Scotland's First University "looks at the foundation of the University and its early years" (*Musa Visitor Guide*). The gallery called Living and Learning "covers student life and experience". The gallery called living and learning "covers student life and experience" (*Musa Visitor Guide*). The gallery called Seeing and Believing "showcases creativity and innovations linked to the University, in fields ranging from the arts and sciences to divinity" (*Musa Visitor Guide*). The gallery called The MUSA Gallery "is about how and why we collect" (*Musa Visitor Guide*).

The positioning of the doors between the galleries adds some restriction to visitors' route of exploration. Visitors will be inclined to follow a linear route through the galleries in increasing order of gallery number.

3.2 Technical provisions at Musa

WiFi is available for both students and visitors at Musa. The curator did, however, say that visitors often have trouble connecting to the WiFi. The curator suggested that it could be possible to provide iPads to visitors. These factors need not play a significant part in the decisions made for this study, however, as evaluation will be carried out under controlled conditions.

4. Software engineering process

4.1 Functional requirements

1) The visualisation should be suitable for a medium-sized display.
2) The visualisation should be suitable for a mobile device.
3) The interface should point visitors to other information/artefacts of potential interest.
4) The interface should provide additional information about selected artefacts.
5) The interface has to support shared interactions.
6) The interface should take an item-centric approach to artefact selection.

4.2 Non-functional requirements

1) The visualisation must be intuitive to use.
2) The interface must not distract users from the actual artefacts.
3) The interface must be visually attractive.

5. Ethics

As part of this project I conducted a study with museum curators to gain feedback about their experience with the prototypes. This involved an audio recording to capture and analyse curators' feedback on the application in more detail. All participants provided informed consent that they may be directly quoted in the report before participating in the study, and they were informed about the methods of data collection i.e. audio recording.

6. Design

There were no constraints placed on the design of the interface besides that it must comply to the functional and nonfunctional requirements. That is, it must, for example, allow the user to find additional information about the artefact that is presently the centre of attention, and it must also allow the user to find artefacts connected to the one that is at the centre of attention. An open-ended ideation process was used, which explored various ideas on how such a collection interface could be designed.

6.1 Sketch-based Ideation Process

The Hexagon Interface

The idea for this sketch came from the drawbars of a drawbar organ. Figure 1 shows the drawbars of an organ. On a drawbar organ, the drawbars would be used to adjust the harmonics that are audible. In this case, the drawbars are used to show the relative strengths of different types of connections between different artefacts. Each drawbar may, for example, represent an historical, geographical or categorical connection, respectively. Moving the drawbars will adjust the relative strengths that the different connections play in deciding the spatial positions of the hexagons in the interface. The centre-most hexagon represents the artefact that is selected for viewing at a given point in time. The other hexagons that are closest to the central hexagon are those that are most closely connected to it. Each hexagon's colour is a categorical visual variable that encodes which of the 4 galleries the artefact belongs to. A fifth colour may be used to denote that an artefact is in the store. Figure 3 shows that drawbar interface.

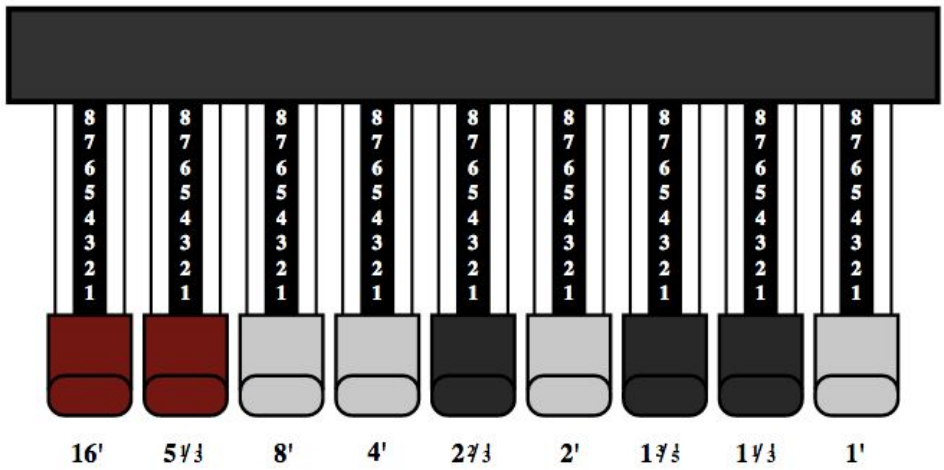


Figure 2. Draw Bars of a Drawbar Organ

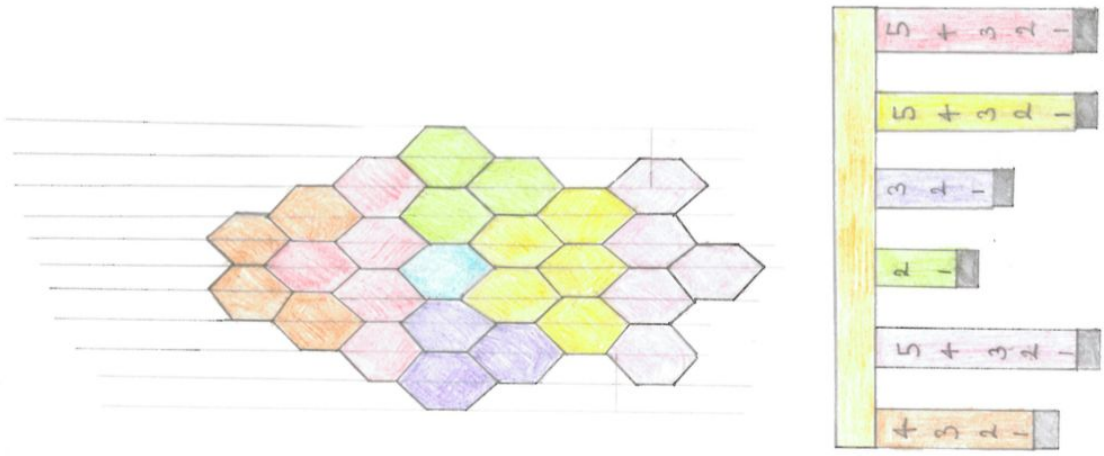


Figure 3. Drawing Connections with Hexagons and Drawbars

The Circle Interface

Every artefact is represented by a different circle. The circles are colour-coded according to the gallery that they belong to. There are 4 colours, each corresponding to one of the 4 galleries. A fifth colour may be used to encode that an artefact is kept in the store. Circles of a particular colour occupy a particular area of the screen. When a particular circle is tapped it is selected. The circles that represent artefacts that are connected to the selected artefact move closer to the selected artefact and the other circles move further away to make room for the connected circles to get close in proximity.

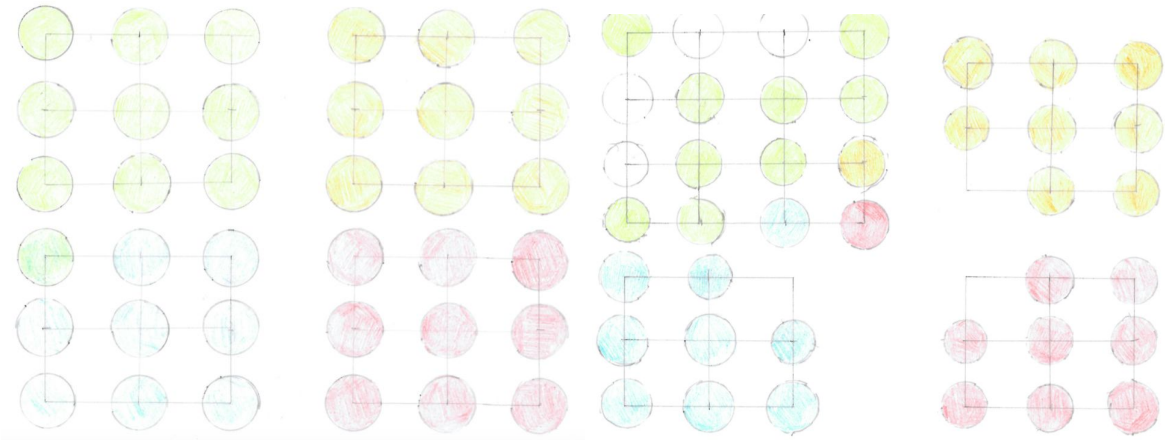


Figure 4. Drawing Connections with Coloured Circles

The Rose Window Interface

The idea for this sketch came from a gothic style of stained glass window used in cathedrals. Figure 4 shows an example of such a window. The outer circle is divided into segments that may be viewed as petals. Each 'petal' can be used to represent an artifact in the museum collection. A petal is coloured according to the gallery or store that it belongs to. The visual variable used to draw connections between artefacts is the colour of a particular petal. The inner circle, or 'pollen' of the flower, represents the selected artefact. The sketch of this interface is shown in Figure 5.

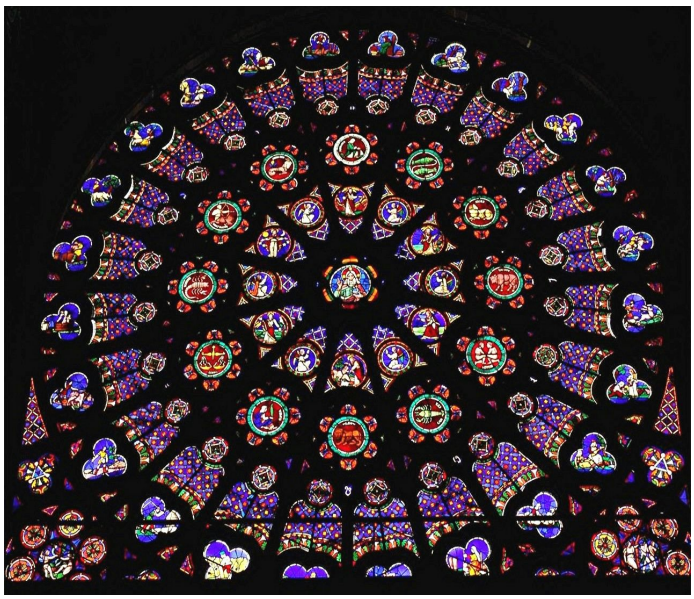


Figure 5. The North Rose of the Abbey of St Denis, Paris (Wikipedia)

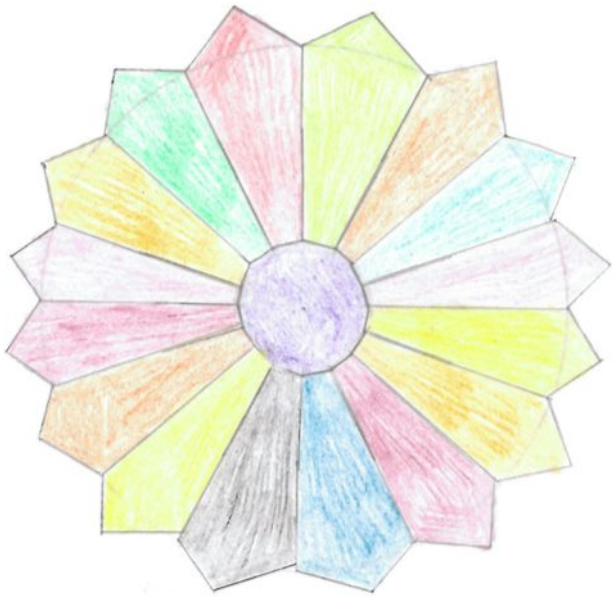


Figure 6. Drawing Connections with Flower Petals

The Organiser Book Interface

An organiser book is a metaphor that many people can relate to. The book would be digital, not paper. Each page of the book contains the picture and information for a different artefact in the collection. The tabs change dynamically. Each tab gives the title of an artefact that is connected to the selected artefact. By clicking on a tab, the user can navigate to the book page for that artefact. The newly selected artefact will have its own new set of tabs for its connected artefacts.



Figure 7. Drawing Connections with an Organiser Book Metaphor

Each sketch carries its own advantages and disadvantages. Looking at the hexagon interface, the colour coding visual variable for the object gallery categorical variable is good for identifying which gallery an artefact belongs to is appropriate. In terms of drawing connections, however, it is only possible to put a maximum of 6 hexagons adjacent to any given hexagon. This limits the number of connections that can be drawn using the position visual variable unless a hexagon may still be considered connected if it is touching a hexagon that is a step further away.

The rose sketch has the advantage that any number of petals can be fitted onto it by reducing the angle that each petal occupies. This means that, taking an item-centric approach, any number of connections can be drawn to the central 'pollen' artefact by adding however many petals are necessary. While this is ideal for making connections to the root node, it is not easily feasible to draw connections between nodes that are not the root node at any given point in time. Colour coding may be used to draw connections by gallery but any other colours would have to be drawn according to position, which is not feasible because only a maximum of two petals can be adjacent.

The major advantage of the circle sketch is that colour and position can both be used as visual variables. This is because by giving circles appropriate sizes any number of circles can be made to surround a given root circle. On the other hand there is the disadvantage that the different colours of circles are meant to be arranged into different corners. There are only 4 corners, meaning that there is only room for 4 galleries to be arranged that way. This is a problem if a fifth colour of circle is added to represent the store because there are not enough corners to support his. It is also questionable whether or not this interface is appropriate for item-centric exploration because a user must select an artefact from the set of all displayed artefacts by tapping it.

The problem of item-centricity is certainly solved by the organiser book sketch. This interface would be inherently item-centric because only one page is displayed at a time and each page contains the information for one single artefact. Item selection by QR code removes the need to narrow down by searching. A book is a familiar concept to almost all people living in normal society so it is guaranteed to be intuitive to use. In terms of disadvantages, however, there is notably no insight given into the nature of a connected artefact before navigating to it besides the title that is given on the tab.

Paper prototypes

Three paper prototypes were produced that could be used for evaluations. The paper prototypes model the three final design concepts. The three final design concepts are the organiser book, the tile and the force-directed graph designs. All of the paper prototypes

were produced to fit within the dimensions of a 9.5 inch iPad display, which is considered a medium-sized mobile display.



Figure 8. Organiser Book Prototype

An organiser book is a mobile device in its own right. A software version of an organiser book is, therefore, a suitable application for a mobile device. Users are pointed to other artefacts of potential interest though the dynamically changing tabs, which provide links to objects connected to the object on the current page of the organiser book. Additional artefact information is provided through an artefact title, image and description on the current page of the organiser book. The prototype takes an item-centric approach to selecting artefacts because the current page of the book is selected by scanning a QR code beside the physical artefact within physical space.

This prototype is also intuitive to use because we can safely assume that the majority of people know how to use a book. An in-situ evaluation would be required to identify whether or not this prototype distracts users from the actual artefacts. Whether it is visually attractive is subjective but ultimately it is a neat and ordered design. We, therefore, see that the organiser book takes into consideration all functional and nonfunctional requirements besides the requirement for support of shared interactions.



Figure 9. Tile Interface Menu



Figure 10. Tile Interface Information Screen

The Windows 8 tile menu on which this prototype was based was designed for a mobile device. We, therefore, assume it to be suitable for a mobile device. The information screen that appears on selection of a tile in the tile menu provides additional information about selected artefacts. The user is pointed to other artefacts of potential interest by the scrolling menu of connected artefacts at the bottom of the information screen. Notably, the prototype does not take an item-centric approach but could be modified to do so by removing the tile menu and limiting the interface to just the information screen. No particular consideration is taken to support of shared interactions.

The fulfillment of the nonfunctional requirements will be identified in the evaluation of this prototype.

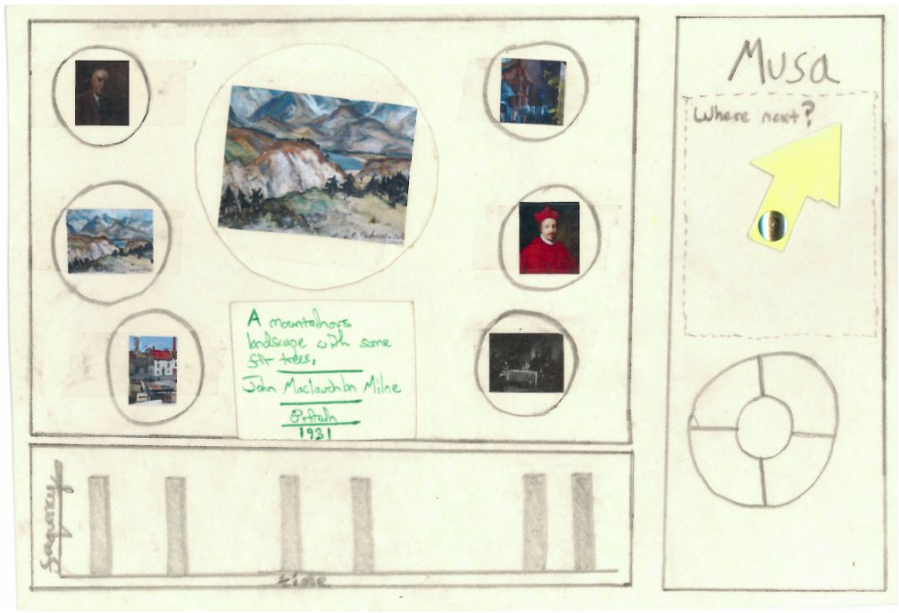


Figure 11. Force-Directed Graph

6.2 Graph-based Interface

A graph is a representation of a set of objects and each artefact in the collection is an object. A graph is, therefore, well-suited to the application of representing artefacts. In a final ideation phase therefore the student, instructed by his supervisor, explored the possibility of showing connections between museum artefacts through a node-link diagram (see Figure 12). We assume the iPad display size of 9.7 inches to define the size of a medium-sized mobile display. To fulfill the functional requirement that the interface should be suitable for a medium-sized mobile display, the sketches were all drawn approximately to those dimensions.

Each vertex of the graph would represent an artefact and each edge would represent a connection between two given artefacts. An artefact may be a painting or a pewter bowl, for example. A connection between two artefacts may be determined by a similarity in two corresponding data attributes in two different rows of the Musa data, where each row represents a different artefact.

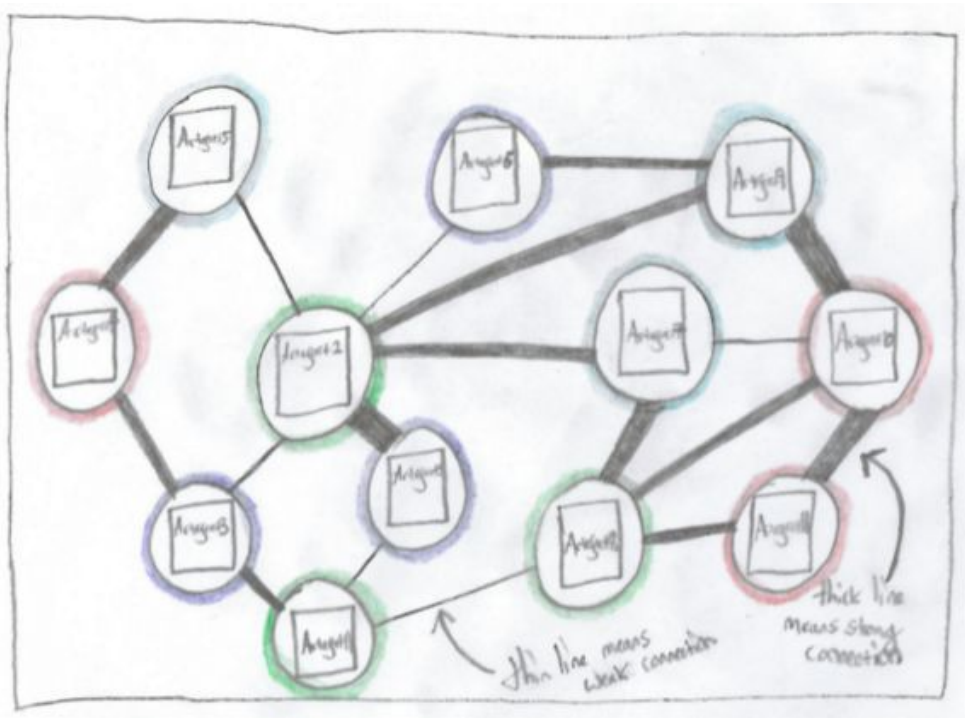


Figure 12. A Force-Directed Graph where the Vertices Represent Artefacts and the Edges Represent Connections between Artefacts

Carrying ideas forward from the previous sketches, the use of colour as a visual variable to denote a particular gallery is used in the force-directed graph. The idea of centering an item and positioning connected artefacts around it is carried forward from the hexagon interface and the flower interface. Another idea drawn from the circle interface is that of the circles rearranging when a particular circle is selected - in the force-directed graph there will be no order in the positioning of the nodes until an artefact is selected. An example use-case would be that a visitor has discovered an artefact at the museum and scanned its QR code. The artefact would then slide into the centre of the interface, enlarged slightly and with descriptions presented in tooltips. This area of the design fulfills the requirements for an item-centric approach and to provide additional information about selected artefacts. Related artefacts would arrange around it with lines as connections.

With the related artefacts arranged around the selected artefact, the visitor could then touch a circle depicting a related artefact. The related artefact would, in turn, slide into the centre, just as the previously selected artefact did and related artefacts now connected to this newly selected one would arrange around it at the peripheries of the viewport.

Interlinked Visualisations

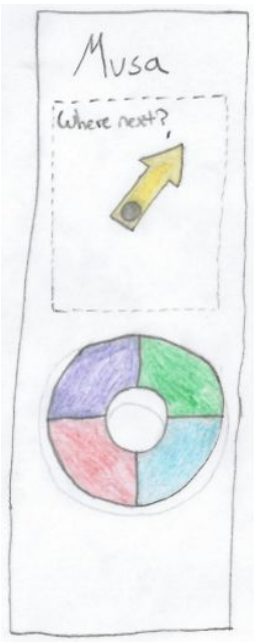
Drawing pointers from It Doesn't Compare to Being There (Rogers, Hinrichs & Quigley, 2016), the full design of the graph-based idea comprises interlinked visualisations. While the

force-directed graph is the main focus of the visualisation, there are is also a sidebar and a timeline. The sidebar contains a compass and a pie chart (see Figure 13).

This last interface shows the most promise to highlight relations of potential artefacts of interest, based on a single "trigger" artefact. It was therefore decided to create a high-fidelity prototype from this final idea.

Sidebar

The compass concept used in the design was conceived by the student. It is based on a metaphor of a real compass but does not always point due north. Instead the direction it points is determined by the location of the gallery containing the most artefacts connected to the selected artefact at a given point in time. The algorithm to determine the direction that the compass should point would be based on the direction that the iPad is facing at a given point in time as well as the location of the gallery containing the largest number of artefacts connected to the artefact that is currently selected.



The pie chart shows the relative proportions of artefacts connected to the currently selected artefact that exist in the 4 different galleries. This idea was conceived by the student. It was criticised by the supervisor, however, who describes the pie chart as *"l'enfant terrible"*, stating that it is a bad type of visualisation for making comparisons. It was agreed that it would suffice
given that there is no critical requirement for accuracy when navigating a museum. The compass and the pie chart are intended to work together in order to fulfill the functional requirement of pointing visitors to other artefacts of potential interest.

Figure 13. Sidebar of Force-Directed Graph Prototype

Timeline

Given that the Musa collection contains 15000 artefacts, the number of vertices displayed in the graph may be overwhelming to the user or simply too great a number to display. The timeline provides filtering functionality. The user may filter the dataset by dragging a 'brush' across the timeline. This allows the user to filter the dataset by the production date of the artefact.

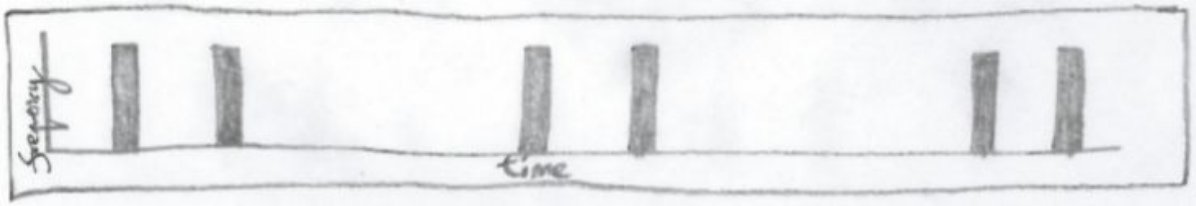


Figure 14. Timeline of Force-Directed Graph Prototype

7. Implementation

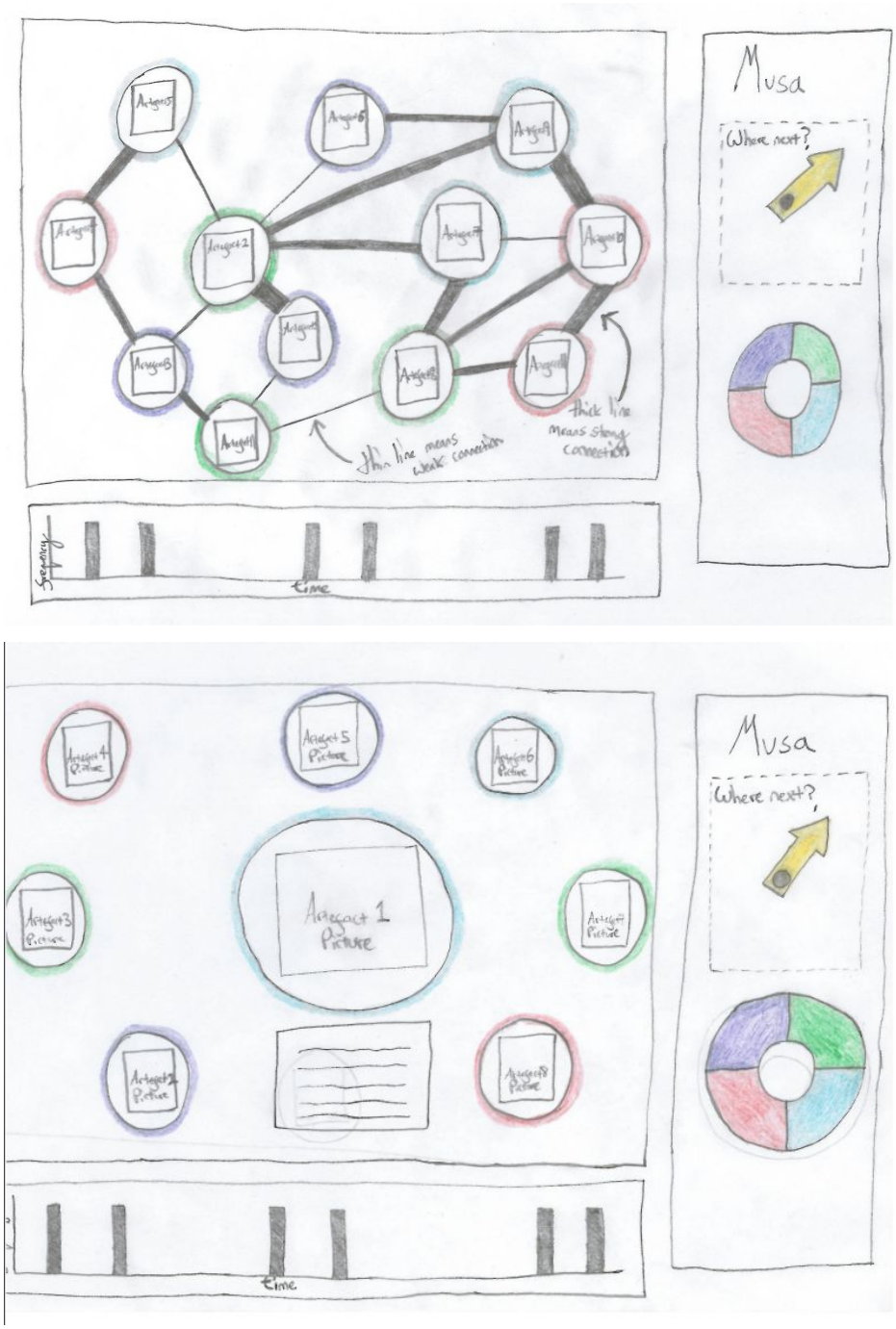


Figure 15. A Complete Force-Directed Graph Visualisation where the Vertices Represent Artefacts and the Edges Represent Connections between Artefacts

There were a number of options open for the implementation language at the start of the project. The project proposal stated that the visualisation should be a tablet interface. It suggested that experience in JavaScript, PHP, D3.js or Objective-C would be necessary. This provided an initial pool of options.

Objective-C was investigated and considered briefly but was discarded on the grounds that would limit device compatibility to Apple devices. Objective-C also carries the problem of how to distribute the software to devices owned by museum visitors. It could be hard to guarantee a reliable internet connection to the visitors' devices and visitors could be discouraged from using the visualisation by perceived complications in its download and installation.

A previous information visualisation SH project for visualising CPU processes (King, 2014) supervised by Dr Graham Kirby had been implemented using Java so Java could have been another option. The student was unaware at the time that Java was a feasible language for implementing visualisations. Otherwise the student would have chosen Java due to his prior experience in it and its ease of debugging in Eclipse.

Viewing server-side implementation as favourable due to the ease of accessibility of a web-based service, PHP and JavaScript became the final contenders. The student had learned JavaScript earlier in the degree in the Advanced Internet Programming module. JavaScript was chosen over PHP because the student had only trivial experience in PHP. JavaScript also offers the convenience of native compatibility with D3.js, which acts in its favour. The use of web programming languages such as JavaScript and D3 satisfies the functional requirement that the visualisation should be suitable for a medium-sized mobile device.

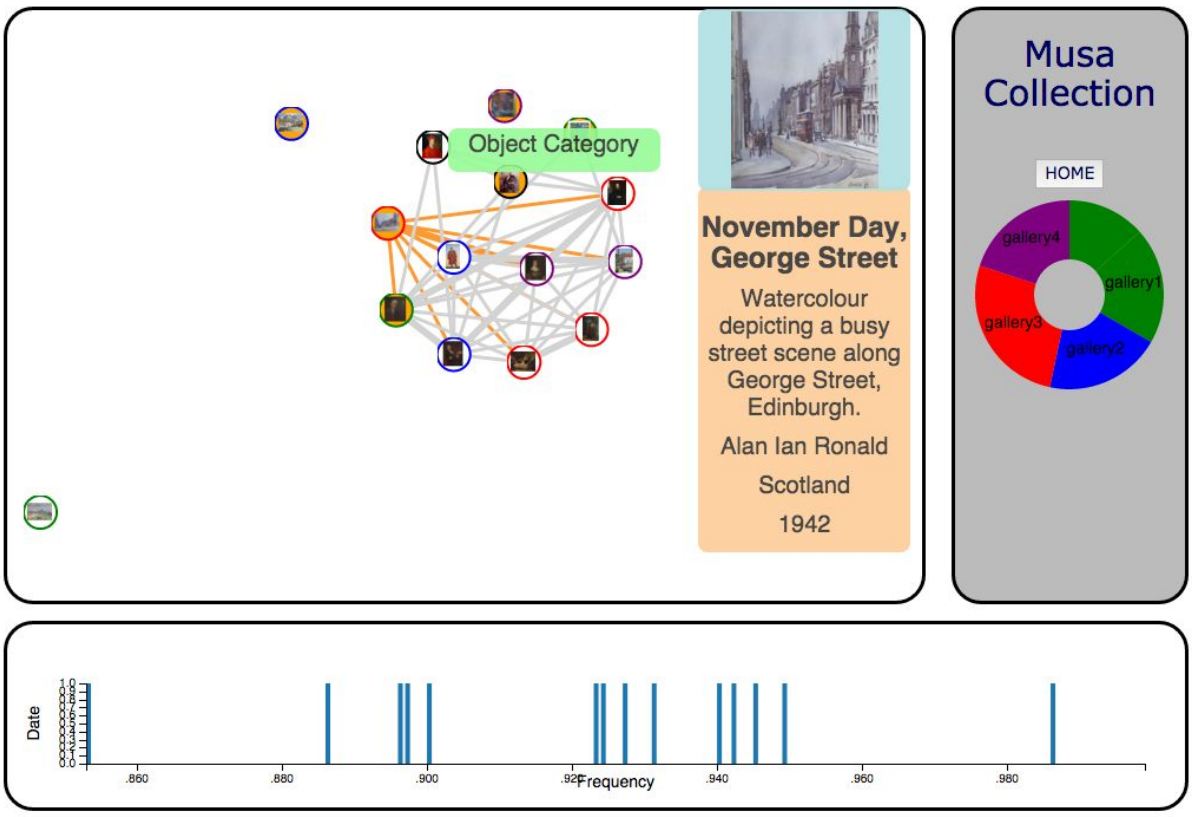


Figure 16. Software Implementation of Force-Directed Graph

The software implementation of the force-directed graph prototype was done using d3.js and dc.js. Consequently, it also involved use of HTML and CSS. d3.js, dc.js and CSS were all new skills for the student. HTML div elements were also a new concept for the student.

There were also a number of additional JS libraries used in this project. There are many JavaScript libraries that support creation of tooltips. Opentip.js is one such example. The actual tooltips used are implemented using HTML divs, however. A D3 transition is used to change the opacity of a particular div in order to reveal it to the user when the appropriate event is fired. This appear an unusual implementation decision. The reason for using this approach was that it appeared to be more elegant implementation than some previous attempts using JavaScript libraries.

D3.js is a Javascript library that may be used to produce information visualisations. It works by manipulating SVG elements in an HTML document via the DOM. The method chaining notation makes it easy to modify parameters of a particular visual element. D3.js had been used for previous museum information visualisations such as It Doesn't Compare to Being There [Rogers]. Given that Dr Uta Hinrichs, the project supervisor, had experience in D3.js, use of D3.js would be additionally appropriate.

DC.js is a Javascript library built using D3.js that allows charts to be created at a higher level of abstraction. Charts it can produce include bar graphs, line charts and pie charts. A chart may be produced by simply creating a new object of the type of that chart and passing data as input in a data structure such as an array.

Crossfilter.js is a Javascript library that may be used to filter tables of data in JavaScript. It may be used in conjunction with DC.js to filter the data that comprises the graph. There are many examples of this having been done in the past, hence the decision to do this in the implementation.

The components of the interface are implemented so that they scale dynamically to the size of the device display. This is done by scaling the interface components relative to the windowWidth property of the JavaScript window object.

Limitations of the implementation

The following table shows the level of completeness of the implementation.

Design feature	**Completeness of implementation**	**Implementation technique / Intended implementation technique**
Force directed graph displays artefacts and connections	Implemented	D3 force layout
Node highlighted on node mouseover	Implemented	D3 events and transitions
Links highlighted on node mouseover	Implemented	D3 events and transitions
Pie chart shows relative proportions of artefacts in galleries	Implemented	DC line chart
Pie chart acts as key to colour coding of galleries	Implemented	DC pie chart
Node centred on node click	Unimplemented	D3 event and D3 transition between 2 different graphs. Each of the two graphs is a different view.
Connected artefacts arranged around selected artefact on node click	Unimplemented	D3 event and D3 transition between 2 different graphs. Each of the two graphs is a different view.
Compass points to physical location of gallery with the most connected artefacts	Unimplemented	New custom-made D3 visualisation in which angle is the visual variable of an arrow shape.
Drag event on timeline causes filtering of artefacts	Partially implemented. Nodes are updated on graph but not the links	DC line chart brush event and crossfilter.
Expand node slightly on mouse over event	Implemented but disabled by a comment in the code	D3 event and D3 transition.

Implementation Challenges

First implementation challenge was getting accustomed to the D3 method chaining syntax. Once adjusted, however, the main challenge was implementing the mechanism by which the root node moves into the centre of the viewport and the connected nodes arrange themselves around the outside. The first attempt at this area of the implementation involved putting a conditional on the x and y attributes that were assigned to the svg elements of the circles in the tick event handler. This did cause the nodes to reposition but proved untenable because the nodes moved sharply, rather than gradually over time. The supervisor suggested that D3 transitions may be a viable solution to this problem. Using transitions on an svg circle's x and y attributes the student was able to cause the root node to move to the centre of the connected nodes to disperse gradually but this caused a problem because the force simulation was still running so gravity would just pull the nodes back into the middle after the transition. The interference of the simulation could be fixed by setting the 'fixed' attribute or a node to true or by stopping the simulation, but this defeated the point of the graph being force-directed.

Another challenge in attempting to implement the node rearrangement transitions was that in the implementation attempts it had been necessary to remove the link lines from the D3 visualisation. The link lines should have been kept and increased in length while the connected nodes arranged themselves outward. The link lines would also have had to be arranged in a circle around the peripheries of the viewport as the nodes were in order to prevent them overlapping with the centred root node. This implementation challenge was not solvable by the student within the time constraints. It could have been possible to implement this by performing transitions on the x and y attributes of the link lines and on the length of the lines.

The link lines caused further implementation challenges when implementing the filtering functionality on the DC timeline. The appears to be filtered correctly and a subset of the original nodes is displayed but the links are removed and the simulation stops. It should be necessary to filter the links correctly too and replace them on the graph when the artefacts have been filtered.

The implementation remains incomplete. Clicking on a vertex does not result in it becoming the root node, moving to the centre of the viewport and the other vertices dispersing to the edges of the viewport. The student can see two possible solutions to this problem. The first solution is to stop the force-directed graph simulation and perform transitions on the x and y coordinates of each individual node in order to position it in the correct location on the graph.

The second solution is to transition between two separate force directed graphs. The first force-directed graph would have the vertices in the starting position and the second would have them in the end position. A previous working example of this technique may be seen at https://vida.io/documents/Aea4ZgQZ88Fx6AZMu .

8. Evaluation/Critical Assessment

The force-directed graph prototype was demonstrated to the Senior Honours project module coordinator on 9 March 2016. The tile interface design was also demonstrated. The module coordinator was Dr Tom Kelsey. Dr Kelsey suggested that an evaluation with the staff at Musa was in order.

8.1 Walkthrough at Musa

A walkthrough was given to the collections curator and the assistant curator at Musa. The walkthrough was given on 23 March 2016. The collections curator was Ms Jessica Burdge. The assistant curator was Ms Rebecca Prentice. Ms Burdge and Ms Prentice both gave feedback on the prototype concepts that were demonstrated to them. The feedback is as follows.

Organiser Book Prototype

Advantages	Disadvantages
1) Straightforward. ["this one is fairly straightforward"] ["for a certain audience it would work well because it is straightforward"] 2) Usability. May be appealing to older, retired people. ["I think definitely usability. I think when we think about our audience and stuff, particularly maybe older retired people who are not used to all the stuff you can do online"] 3) Jessica and Rebecca like the idea. ["We both like this idea [referring to book idea]"]	1) Frustrating that there are only 4 or 5 related artefacts in the tabs. ["On a negative, I am assuming it only shows you 4 or 5 related things and it might be frustrating actually if you are someone who knows what you are looking for."]

Tile Prototype

Advantages	Disadvantages
1) Straightforward. There are tiles. You click one. ["like that one [organiser] it is a quite straightforward interface, people would understand right away how to interact with that"] 2) Bottom menu good.	1) Tile interface is not easy to filter. ["how is the front page (tiles) generated based on what tiles are here?"] 2) Not item-centric.

3) Good size of menu.	

Force Directed Graph Prototype

Advantages	Disadvantages
1) Scope for less rigid interaction. ["Seems a bit more limitless compared to the first one."] 2) Time and location filters are good. 3) Instructions not included. Is it really intuitive? Jessica thinks that it would be intuitive enough for her. 4) The compass is appealing to the curators. ["I do like the idea of it. It's kind of like making your own journey rather than the prescriptive 'you must walk round this gallery'"]	1) Too many lines. ["Seeing all the lines is really confusing. It looks all mathematical."] 2) Looks mathematical. Looks scientific. Not inclusive to people who are not scientific. Many visitors may be looking for something a bit more informal. ["Seeing all the lines is really confusing. It looks all mathematical."] 3) Nodes look small so can't see what they are. 4) For foreign languages, would the book be easier to translate? (No! Translation is the same for all of them.) 5) Not actually a big museum so compass may be obsolete. 6) Too many connection lines. ["is there some way of not having all the objects linked because there are just too many of them"] 7) Unintuitive. ["how do they get onto the screen where the artefact is picked out?"] 8) Too many nodes to display on the graph at the same time. ["how do they get onto the screen where the artefact is picked out?"] ["would those nodes [before selection] show everything in the collection?"] 9) Line thickness may not be the best way to encode connectedness. ["the thickness of the lines might be too subtle for some people to realise that that is why the lines are different thicknesses"] 10) The compass may be obsolete. ["it's difficult because it's not really a

	very big museum”]

Overall

Advantages	**Disadvantages**
“Trying to think how much time people would be willing to invest, we want to hold their attention on the objects first and foremost. How much time they will actually spend online when they are actually in the museum.”	Drawing connection between Musa data attributes may not be the best way to draw connections. “So I suppose it would be trying to work out how to draw a connection to something else.”
“I don’t think there is need for a whole redesign.”	

8.2 Adding Perspective

Ms Prentice had the idea of adding a new concept to the force-directed graph. It is a concept of perspective. The idea extends that of using the thickness of lines to denote connectedness and aims to solve the problem that “it is not really grabbing you and pulling you in”. In adding perspective, circle size would be used denote connectedness in addition to line thickness. Circle size could vary according to distance of either the physical artefact from the physical position of the user or alternatively according to how strongly connected the artefact denoted by the circle is to the selected artefact.

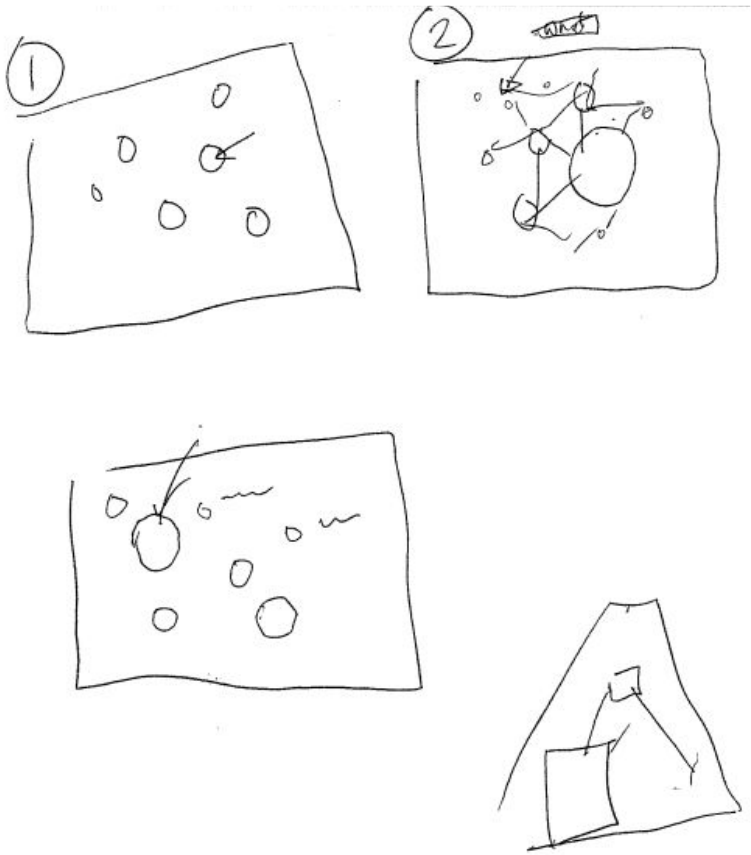


Figure 17. Sketch of the Perspective Concept by Ms Prentice

Ms Prentice suggested that this concept would draw people into the visualisation.

8.3 Analysis and Iterative Re-Design

The feedback obtained in the evaluation should be treated with caution. Some of the feedback they gave was based on premises that were not true. One example of such a comment for the book interface is that it is 'frustrating that there are only 4 or 5 related artefacts in the tabs'. Responses such as this could have been mitigated by correcting the curators on their assumption that the state of the tabs was static and not dynamic. The explanation of the interfaces could also have been prepared better to begin with.

Analysing their feedback further on a meta level, we can see that many of their comments such as 'looks scientific' and 'looks mathematical' are aesthetic, and not functional. It is appropriate that aesthetic comments should be taken into account, however, because at the start of the project Prof Aaron Quigley stated that the interface should 'not look like it was drawn by a child'. This was non-functional requirement placed on the aesthetic of the graph.

Organiser Book Prototype

Both of the advantages given for the organiser book interface suggest that it is an intuitive design. The second of the two suggests that it would even be intuitive to a non-technical

demographic. This puts the design in a positive light. Further still, the only disadvantage given is unfounded because, though there is only space for 4 or 5 connected artefact tabs on the paper prototype, in a software implementation the size of the tabs could be scaled to allow for more than 4 or 5 to fit in the space. Even then, it is good that this potential disadvantage was highlighted by the curators because it emphasises the significance of doing this in the software implementation.

Tile Prototype

The advantages given for the tile prototype would suggest that it would be a good idea to keep the bottom scrolling menu the same in any redesign. The tiles themselves were commended for their usability but criticised for not providing an item-centric approach to interaction. The curators also pointed out that the tile interface may not be easy to filter and that there would not necessarily be space for every artefact to be represented by a tile. The criticism of the tiles suggests that an iterative redesign of this interface should not actually have a tile menu but just an information screen that is navigated via the scrolling menu at the bottom. Figure 18 shows a sketch of a possible redesign of the the tile interface. It comprises the artefact picture at the top with an artefact description below it and a scrolling menu of connected artefacts at the bottom. The artefact picture and description correspond to the artefact that the user is looking at and has scanned the QR code of within the museum galleries.

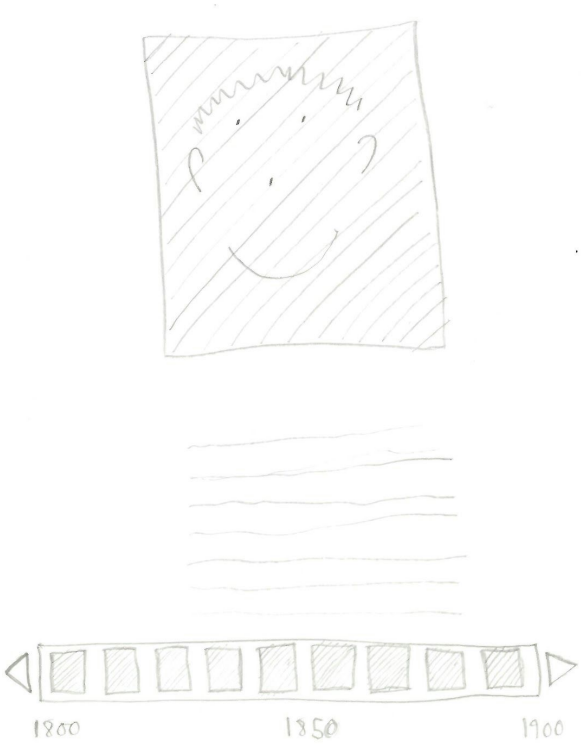


Figure 18. A Redesign of the Tile Interface

Force Directed Graph Prototype

The feedback on the force directed graph prototype would suggest that the time and location filters, and the compass are positive features. The compass may be considered obsolete within the design because the curators said that the museum may be too small to require the functionality that it provides. The compass is, however, the solution to a key functional requirement, namely drawing routes of exploration through the galleries, so the compass should cannot be removed for that reason.

A number of properties of the force directed graph component of the design were criticised. Problems such as the 'mathematical' or 'scientific' look could possibly be fixed by restyling the graph to look more accessible. The 'too small nodes' could be fixed by increasing their size or even just reinstating the functionality that causes the size of nodes to be increased slightly on hover over. The disadvantage of there being 'too many' connection lines could be fixed by only displaying a subset of the connection lines such as the top 15 strongest connections. Notably, these criticisms are of individual characteristics of the graph and not of the graph itself so the graph should still remain a valid design provided that some minor modifications are made.

There are 15000 artefacts in the collection and selection from within that must be item centric. A solution to this problem is to have two separate force-directed graphs and transition between them. The first force directed graph would contain the complete set of vertices for all the artefacts, while the second would contain only the selected route node vertex and its connected artefact vertices. The root node would be displayed in the centre of the viewport and the connected nodes around the peripheries. The transition from the first force-directed graph to the second force-directed graph would be triggered by scanning the QR code of the artefact that is to become the root node. This removes the necessity for the user to find a proverbial needle in a haystack when looking for an artefact amongst all the vertices.

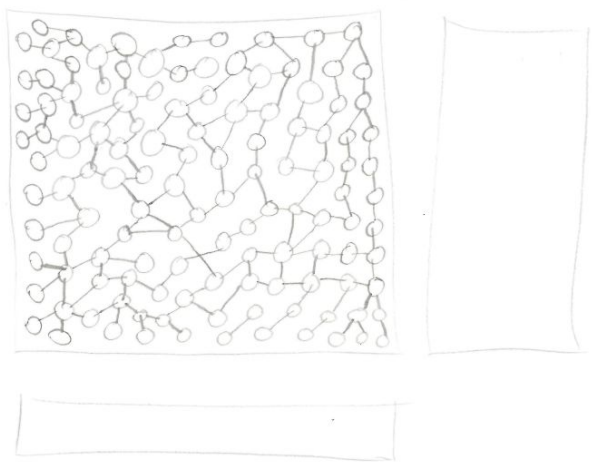


Figure 19. 15000 vertices at once

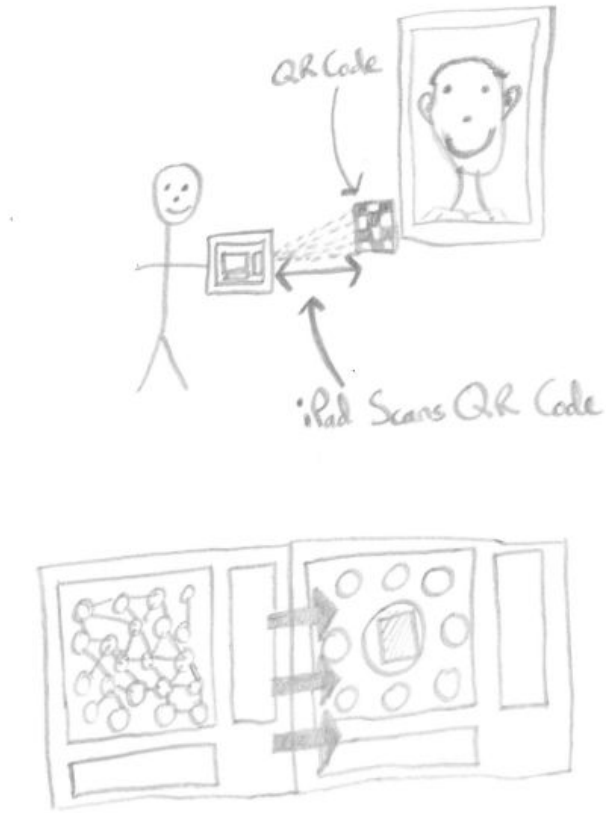


Figure 20. User scans QR code

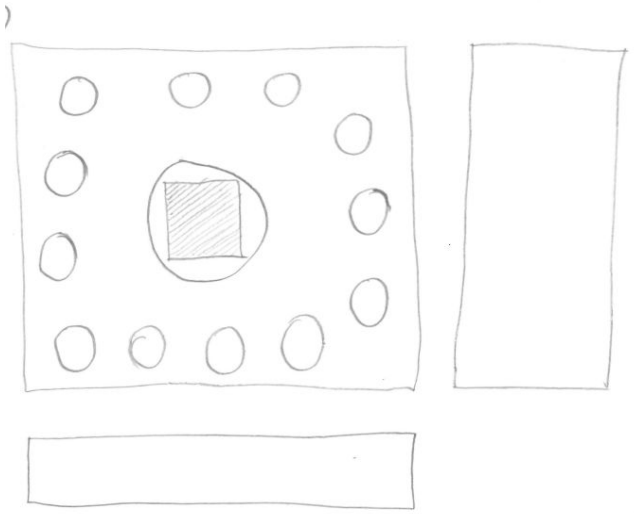


Figure 21. Root node is selected and only connected nodes are displayed

8.4 Conclusions from the evaluation

We can conclude that all of the paper prototypes carry positive and negative features. The organiser book prototype is arguably the best solution to the problem because it only carries 2 usability-based advantages and only one unfounded disadvantage. In retrospect, it may, therefore, have been best to implement a software version of that prototype. This could have been possible if the paper prototype evaluation had been carried out prior to the software prototype implementation.

The tile interface prototype requires a significant redesign that removes the tile menu, leaving only the description page and scrolling menu. This was shown in Figure 18.

The force directed graph prototype requires minor aesthetic adjustment in order to make it appeal to people who are not scientific. It also requires that the node expansion on mouse over functionality should be reinstated and that only a limited number of the selected nodes should be displayed. A transition between 2 different graph views should also be implemented, where the first view contains vertices for all the artefacts and the second has the root node centred and its connected nodes arranged around it.

9. Conclusion

Multiple designs have been ideated, sketched and conceptualised. These designs make good progress towards solving problems such as providing visitors with additional artefact information, helping visitors draw routes of exploration around the galleries and drawing connections between artefacts. The software implementation, meanwhile, makes progress towards making some of these designs a reality.

The evaluation with the Musa curators generated valuable feedback that could be used to improve the design concepts. Consequently, we can resolve to modify the force-directed graph prototype to include only a subset of the connection lines, to improve the styling to make it more appealing to non-scientific people and to add perspective to draw visitors' attention towards artefacts that are more strongly connected to the root artefact. It was also identified that the organiser book design may be a more favourable solution to the problem than the force directed graph because it is intuitive and accessible, and yielded no significant disadvantages in the evaluation.

This study is not complete, however. The reason for this is that the implementation of the software prototype is incomplete and an in-situ user study has not been performed. There was also no consideration taken towards fulfilling the design requirement of supporting shared interactions. This provides fuel for future work in completing and evaluating the

force-directed graph interface as well as in implementing and evaluating the organiser book interface.

10. References

Hinrichs, U., Schmidt, H., & Carpendale, S. (2008). EMDialog: Bringing Information Visualization into the Museum. *IEEE Trans. Visual. Comput. Graphics*, *14*(6), 1181-1188. http://dx.doi.org/10.1109/tvcg.2008.127

King, I. (2014). *Understanding Thread Interactions* (Undergraduate). St Andrews University.

Musa Visitor Guide. (2016). St Andrews.

Rogers, K., Hinrichs, U., & Quigley, A. (2016). It Doesn't Compare to Being There: In-Situ vs. Remote Exploration of Museum Collections. Retrieved from http://searchisover.org/papers/rogers.pdf

Winkler, C., Lochtefeld, M., Dobbelstein, D., Kruger, A., & Rukzio, E. (2014). SurfacePhone. *CHI 2014*. Retrieved from http://dl.acm.org/citation.cfm?id=2557075